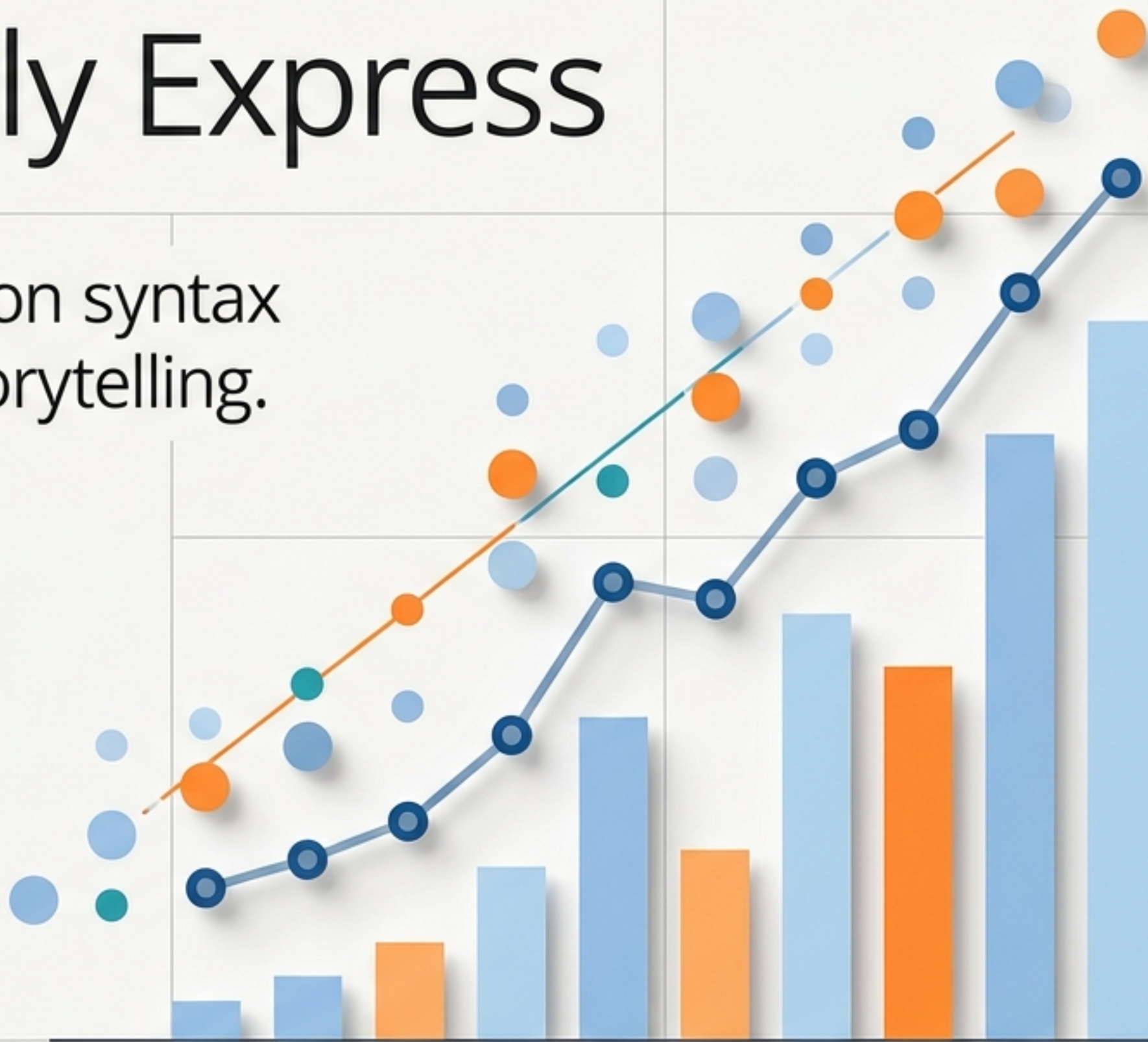


Mastering Plotly Express

A visual guide mapping Python syntax directly to interactive data storytelling.



The Raw Materials: Understanding Our Input Data

student_data (95 rows × 10 columns)

id	class	study_hours	test_score	week	attendance_rate	gender	hostel
0	Freshman	5	78	1	90	Male	True
1	Freshman	3	65	1	80	Female	False
2	Freshman	4	70	1	85	Male	True
...	...	...	...	...	...	...	...
94	Senior	6	84	6	91	Male	False

enrollment_data

Year	Programming	Digital Marketing
2015	589	734
2016	862	963
2017	1085	1157
2018	1150	1404
...	...	...
2024	2442	2338

Mapping Multidimensional Relationships



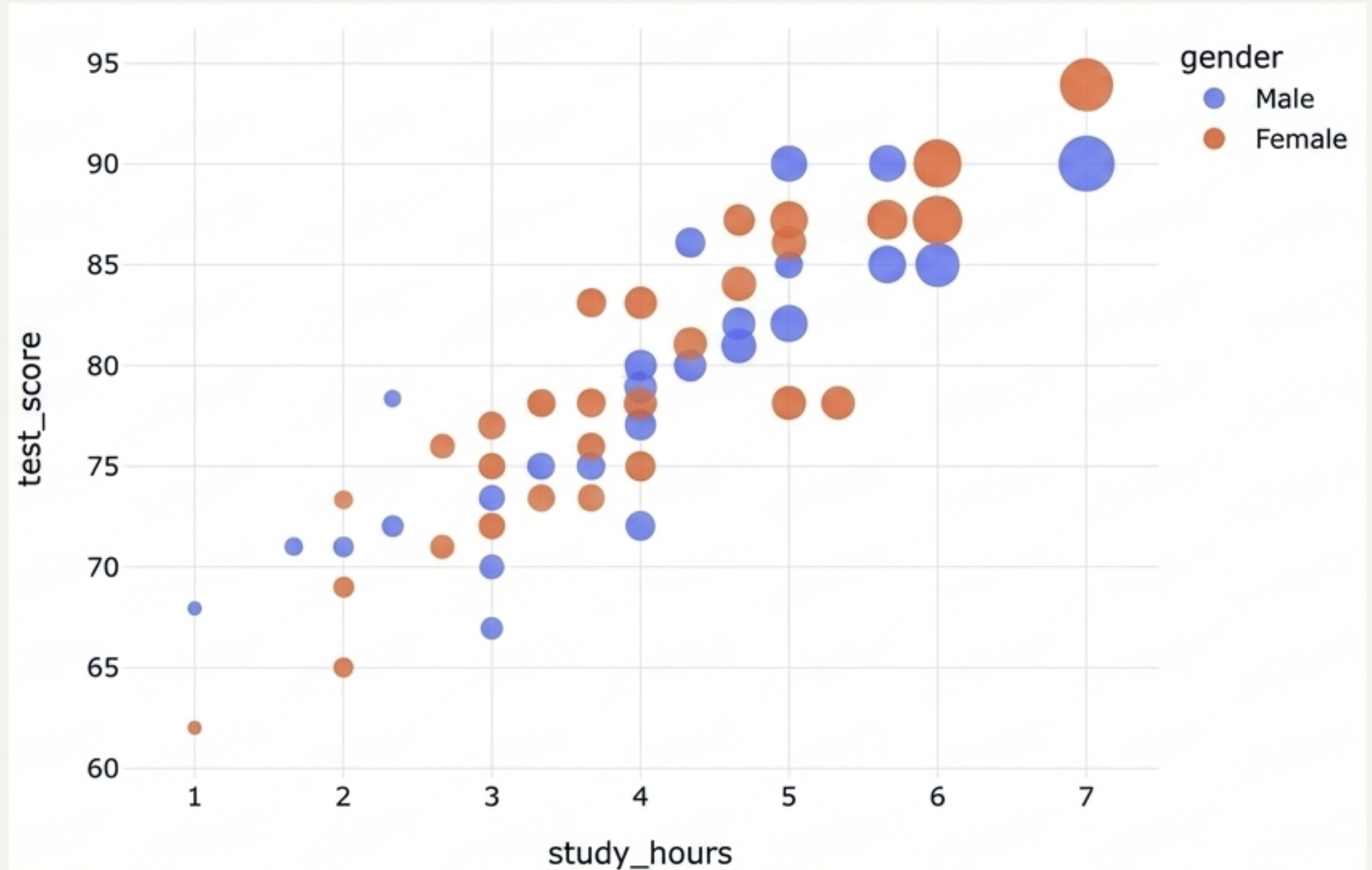
Concept

Visualizing correlation across up to five dimensions simultaneously without cluttering the canvas.

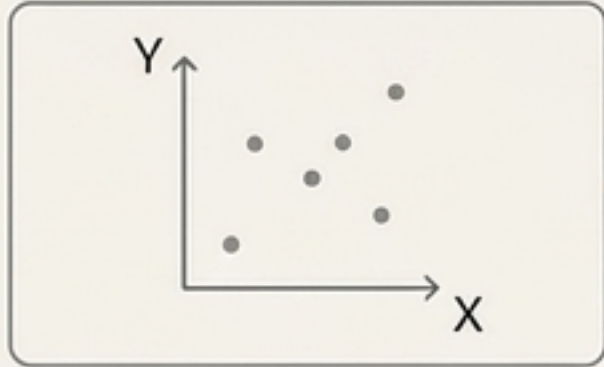


Current View

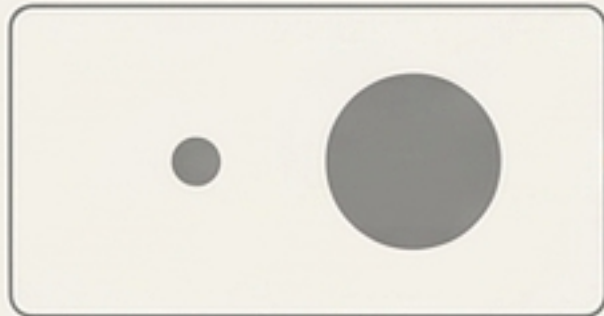
Evaluating how study_hours impact test_score, segmented by student demographics and time.



Anatomy of a Plotly Scatter Plot



Defines the spatial coordinate position of each data point on the canvas.



Uses a numerical column to scale the marker area relative to the value.

```
fig = px.scatter(student,  
x = 'study_hours',  
y = 'test_score',  
color = 'gender',  
size = 'week',  
hover_data = ['hostel', 'class_level']  
)  
fig.show()
```

Legend
● Male
● Female

Automatically assigns distinct hues to categorical groups.

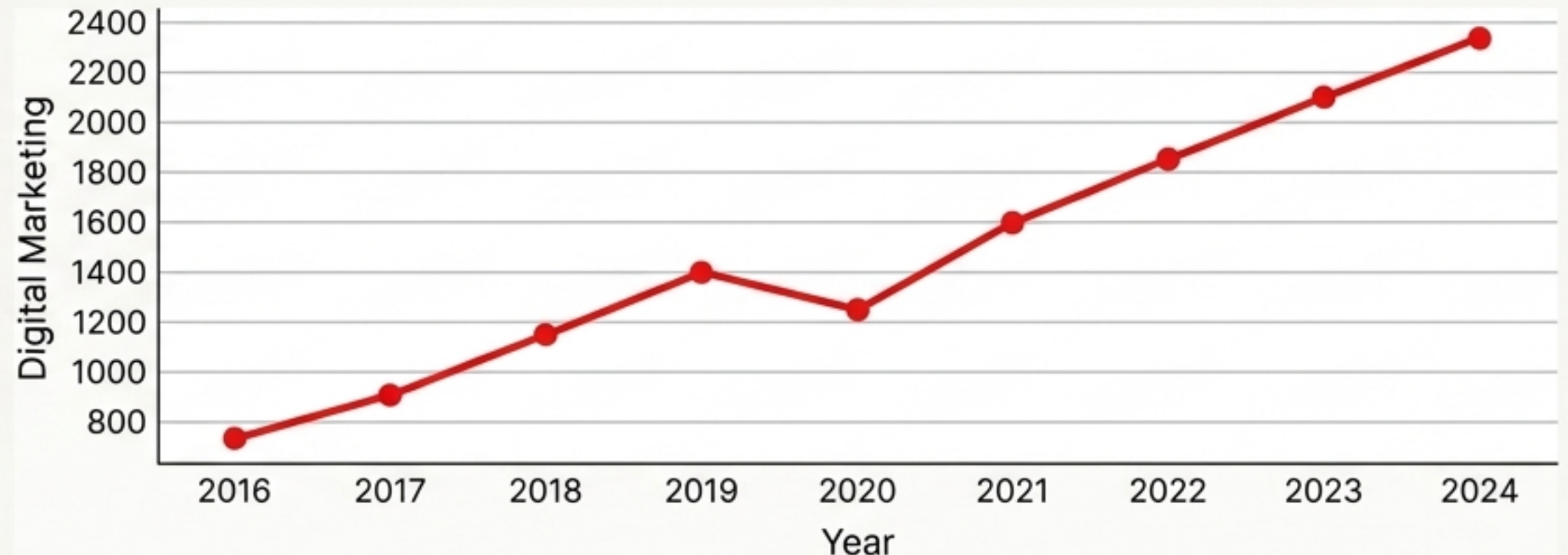
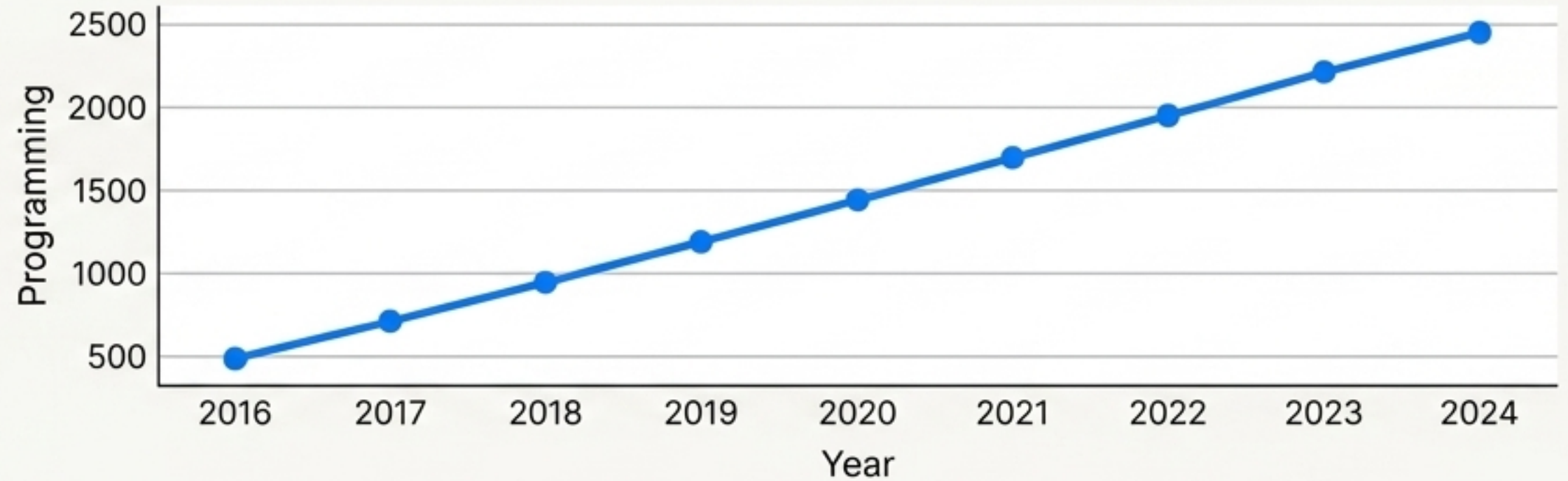
```
study_hours: 5  
test_score: 78  
hostel: True  
class_level: Freshman
```

Injects additional data columns into the interactive tooltip without altering the geometric plot.

Tracking Progression Over Time

Concept

Connecting distinct data points across a continuous sequence (Year) to reveal upward or downward trajectories.

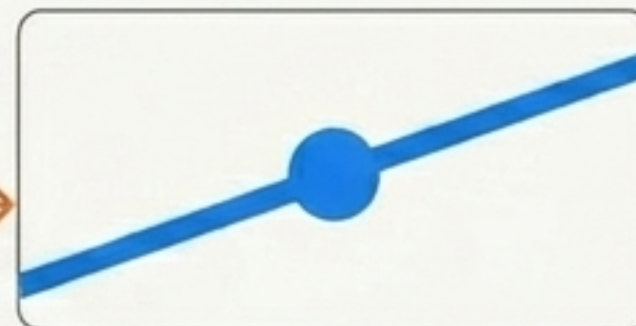


Anatomy of Base Plots and Trace Updates

Base Plot

```
fig = px.line(enrollment_data, x='Year',  
              y='Programming', markers=True)
```

Establishes the continuous line geometry.



Forces Plotly to render explicit data points on top of the continuous line.

Trace Updates

```
fig = px.line(enrollment_data,  
              x='Year', y='Digital Marketing',  
              markers=True)  
fig.update_traces(line_color='red')  
fig.show()
```



`update_traces` acts as a powerful override method to manually customize styling elements post-creation, before the figure is rendered.

Understanding Data Density and Distribution



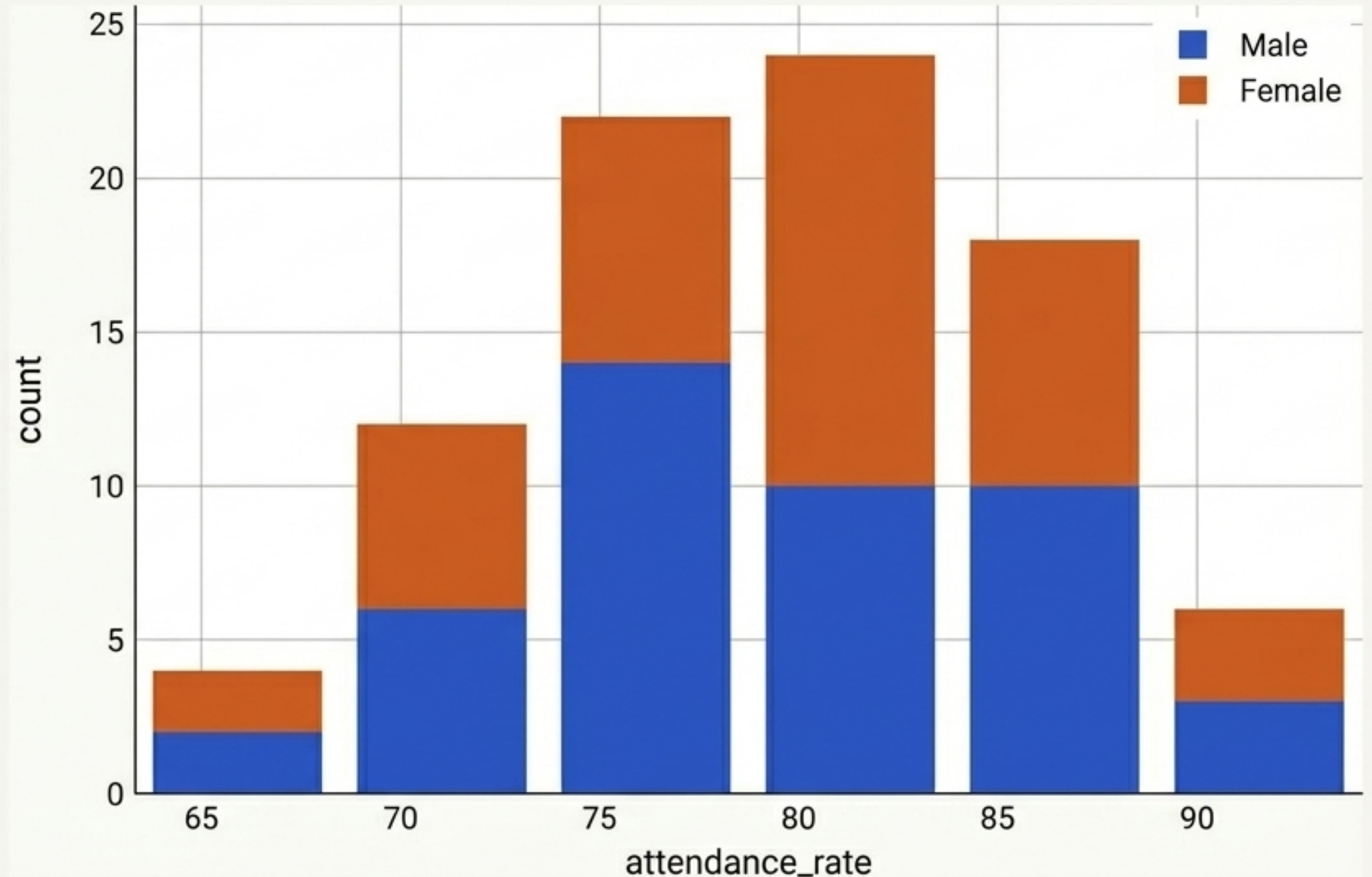
Concept

Grouping continuous data into discrete 'bins' to count frequency and identify where data clusters.



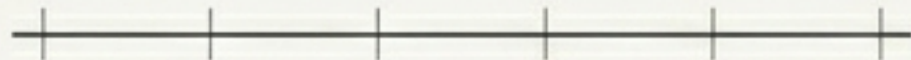
Current View

Visualizing the frequency of student attendance scores, segmented internally by gender to reveal sub-distributions.



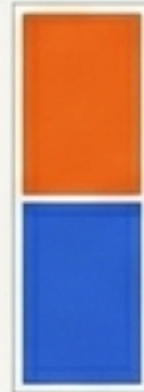
Anatomy of Automatic Aggregation

```
fig = px.histogram(student, x='attendance_rate', color='gender', nbins=10)  
fig.show()
```



attendance_rate

Histograms require only one continuous variable. Plotly automatically calculates the Y-axis as a raw count of occurrences.

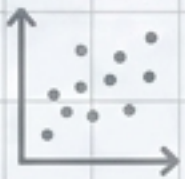


Auto-stacks the categorical variables within the same bin, providing a sub-distribution without overlapping.



Dictates the granularity of the distribution by forcing the data into a maximum of 10 discrete intervals.

The Plotly Express Cheat Sheet

	 Scatter Plot (px.scatter)	 Line Plot (px.line)	 Histogram (px.histogram)
Primary Use	Correlation across multiple dimensions.	Continuous tracking of trends.	Frequency distribution and data clustering.
Data Requirement	Two continuous variables (X, Y).	One continuous axis (usually Time), one metric.	A single continuous variable (Auto-calculates Y).
Signature Parameters	size (numerical scaling) hover_data (tooltip expansion)	markers=True (node visibility) update_traces (post-creation styling)	nbins (granularity control) Auto-stacking via color argument